

Cryptography and Network Security Chapter 2

Fourth Edition
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Chapter 2 – Classical Encryption Techniques

Many savages at the present day regard their names as vital parts of themselves, and therefore take great pains to conceal their real names, lest these should give to evil-disposed persons a handle by which to injure their owners.

—The Golden Bough, Sir James George Frazer

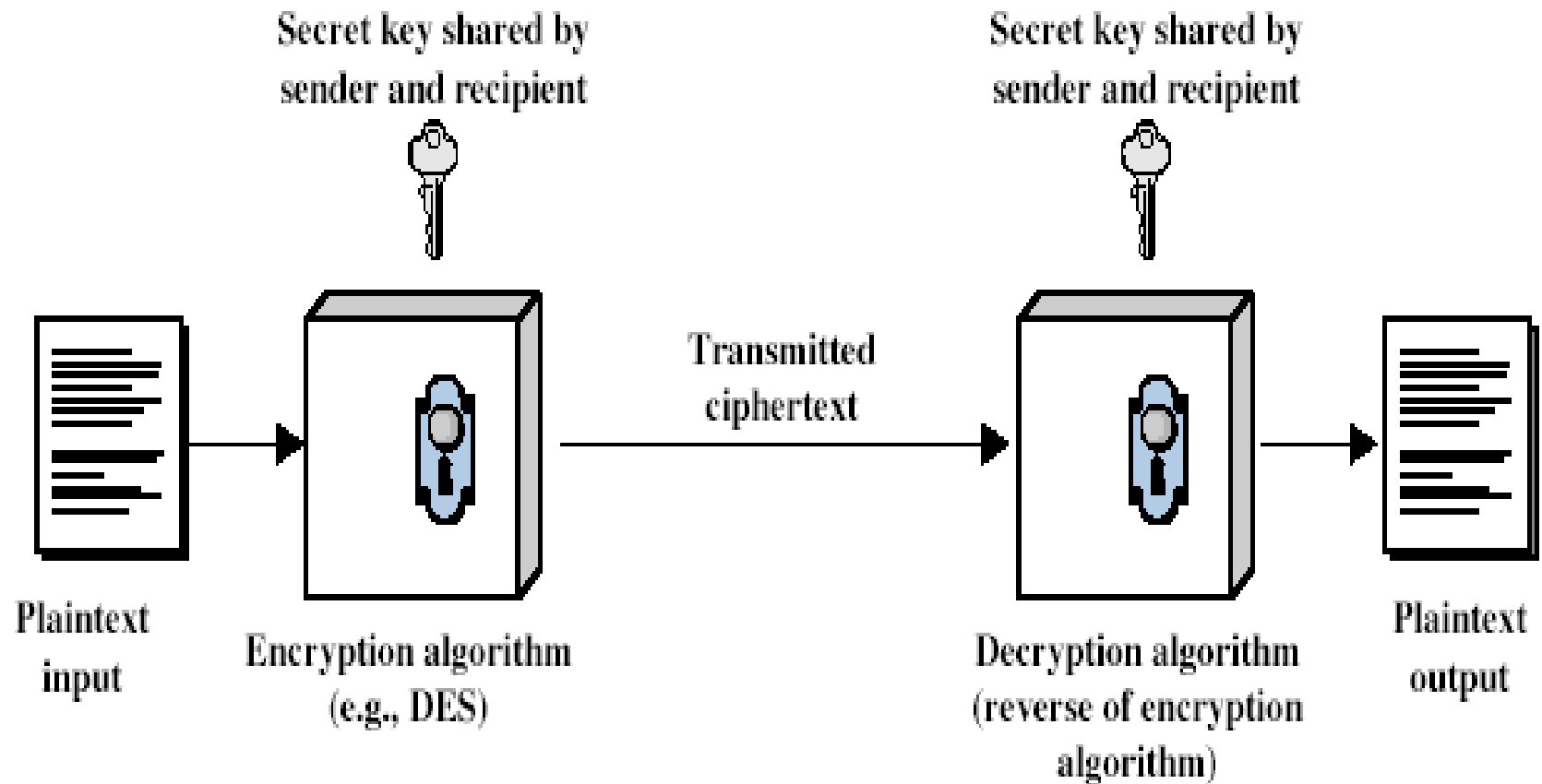
Symmetric Encryption

- or conventional / private-key / single-key
 - sender and recipient share a common key
 - all classical encryption algorithms are private-key
 - was only type prior to invention of public-key in 1970's
 - and by far most widely used
- 
- The bottom right corner of the slide features several decorative concentric circles in a lighter shade of purple, resembling ripples in water. There are three distinct sets of these circles, with the largest one being the most prominent.

Some Basic Terminology

- **plaintext** - original message
- **ciphertext** - coded message
- **cipher** - algorithm for transforming plaintext to ciphertext
- **key** - info used in cipher known only to sender/receiver
- **encipher (encrypt)** - converting plaintext to ciphertext
- **decipher (decrypt)** - recovering ciphertext from plaintext
- **cryptography** - study of encryption principles/methods
- **cryptanalysis (codebreaking)** - study of principles/methods of deciphering ciphertext *without* knowing key
- **cryptology** - field of both cryptography and cryptanalysis

Symmetric Cipher Model



Requirements

- two requirements for secure use of symmetric encryption:
 - a strong encryption algorithm
 - a secret key known only to sender / receiver
- mathematically have:
$$Y = E_K(X)$$
$$X = D_K(Y)$$
- assume encryption algorithm is known
- implies a secure channel to distribute key

Cryptography

- characterize cryptographic system by:
 - type of encryption operations used
 - substitution / transposition / product
 - number of keys used
 - single-key or private / two-key or public
 - way in which plaintext is processed
 - block / stream



Cryptanalysis

- objective to recover key not just message
- general approaches:
 - cryptanalytic attack
 - brute-force attack

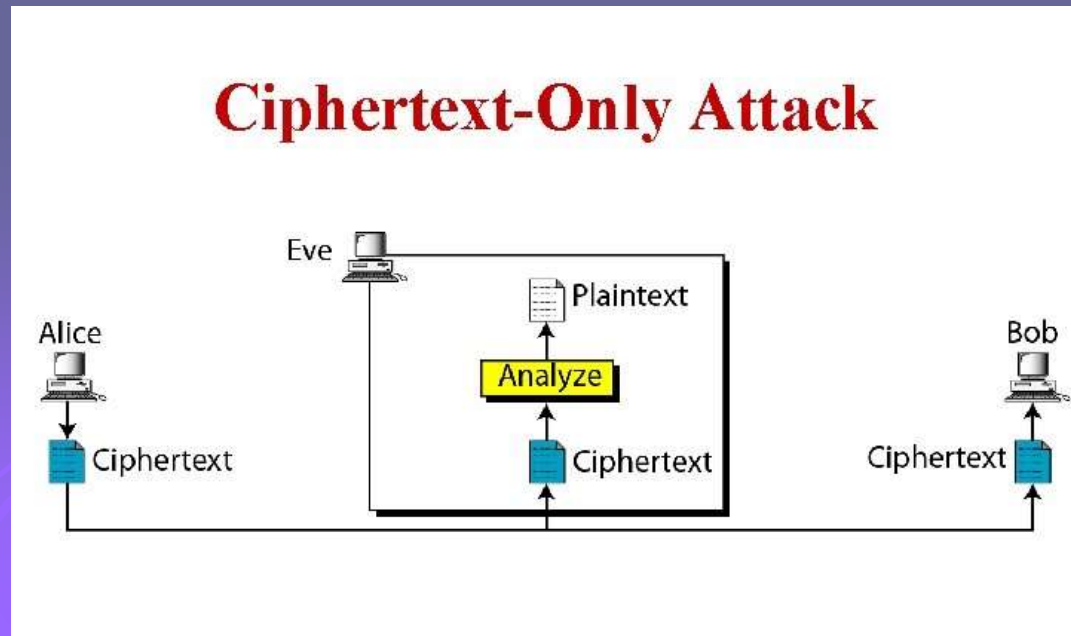


Type of Attack	Known to Cryptanalyst
Cipher text only	• Encryption algorithm • Cipher text
Known plaintext	• Encryption algorithm • Cipher text • One or more plaintext-cipher text pairs formed with the secret key
Chosen plaintext	• Encryption algorithm • Cipher text • Plaintext message chosen by cryptanalyst, together with its corresponding cipher text generated with the secret key
Chosen ciphertext	• Encryption algorithm • Cipher text • Purported cipher text chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen text	• Encryption algorithm • Cipher text • Plaintext message chosen by cryptanalyst, together with its corresponding cipher text generated with the secret key • Purported cipher text chosen by cryptanalyst, together

Cryptanalytic Attacks

➤ ciphertext only

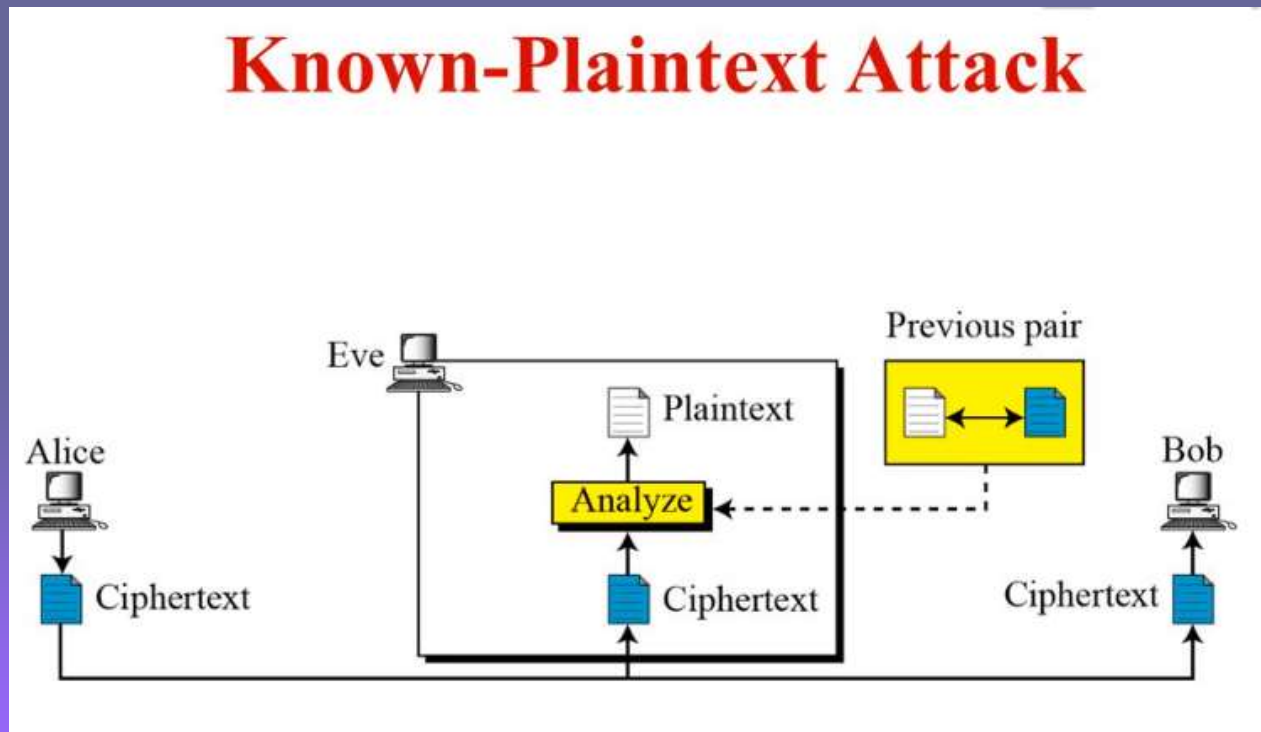
- only know algorithm & ciphertext, is statistical, know or can identify plaintext
- In cryptography, a ciphertext-only attack (COA) or known ciphertext attack is **an attack model for cryptanalysis where the attacker is assumed to have access only to a set of ciphertexts**. The attack is completely successful if the corresponding plaintexts can be deduced (extracted) or, even better, the key.



Cryptanalytic Attacks

➤ known plaintext

- know/suspect plaintext & ciphertext

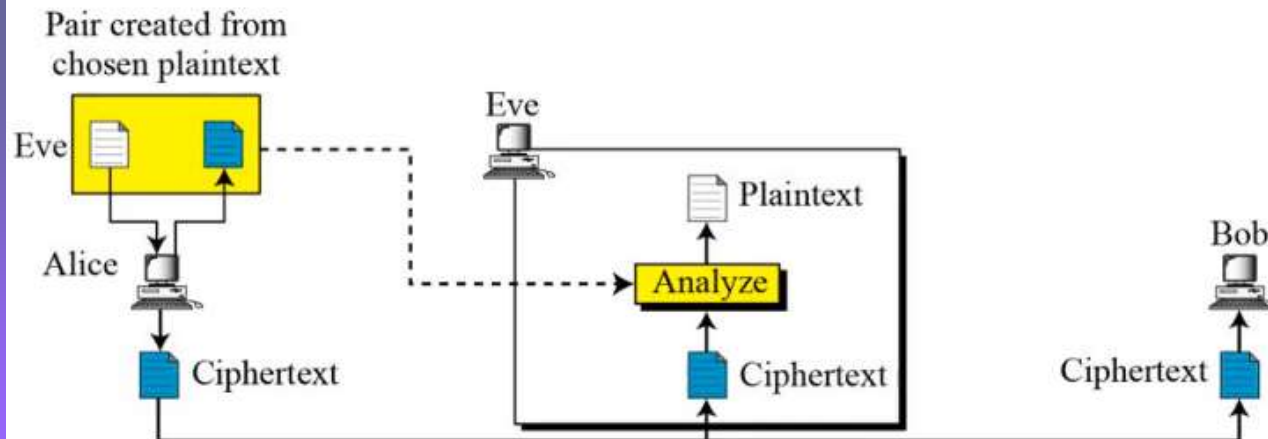


Cryptanalytic Attacks

➤ chosen plaintext

- select plaintext and obtain ciphertext

Chosen-Plaintext Attack



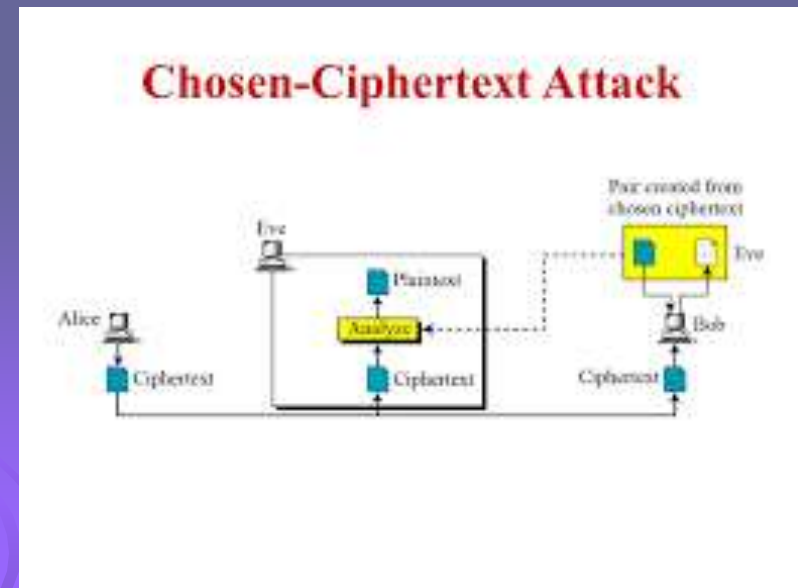
Cryptanalytic Attacks

➤ chosen ciphertext

- select ciphertext and obtain plaintext
- The attacker has capability to make the victim (who obviously knows the secret key) **decrypt any ciphertext and send him back the result**. ... By analysing the chosen ciphertext and the corresponding received plaintext, the intruder tries to guess the secret key which has been used by the victim.

➤ chosen text

- select plaintext or ciphertext to en/decrypt



More Definitions

➤ **unconditional security**

- no matter how much computer power or time is available, the cipher cannot be broken since the ciphertext provides insufficient information to uniquely determine the corresponding plaintext

➤ **computational security**

- given limited computing resources (eg time needed for calculations is greater than age of universe), the cipher cannot be broken

Brute Force Search

- always possible to simply try every key
- most basic attack, proportional to key size
- assume either know / recognise plaintext

Key Size (bits)	Number of Alternative Keys	Time required at 1 decryption/ μ s	Time required at 10^6 decryptions/ μ s
32	$2^{32} = 4.3 \times 10^9$	$2^{31} \mu\text{s} = 35.8 \text{ minutes}$	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	$2^{55} \mu\text{s} = 1142 \text{ years}$	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	$2^{127} \mu\text{s} = 5.4 \times 10^{24} \text{ years}$	$5.4 \times 10^{18} \text{ years}$
168	$2^{168} = 3.7 \times 10^{50}$	$2^{167} \mu\text{s} = 5.9 \times 10^{36} \text{ years}$	$5.9 \times 10^{30} \text{ years}$
26 characters (permutation)	$26! = 4 \times 10^{26}$	$2 \times 10^{26} \mu\text{s} = 6.4 \times 10^{12} \text{ years}$	$6.4 \times 10^6 \text{ years}$

Classical Substitution Ciphers

- where letters of plaintext are replaced by other letters or by numbers or symbols
- or if plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns



Caesar Cipher

- earliest known substitution cipher
- by Julius Caesar
- first attested use in military affairs
- replaces each letter by 3rd letter on
- example:

meet me after the toga party

PHHW PH DIWHU WKH WRJD SDUWB



Caesar Cipher

- can define transformation as:

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

- mathematically give each letter a number

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

- then have Caesar cipher as:

$$c = E(p) = (p + k) \bmod (26)$$

$$p = D(c) = (c - k) \bmod (26)$$

Cryptanalysis of Caesar Cipher

- only have 26 possible ciphers
 - A maps to A,B,..Z
- could simply try each in turn
- a **brute force search**
- given ciphertext, just try all shifts of letters
- do need to recognize when have plaintext
- eg. break ciphertext "GCUA VQ DTGCM"

Brute-Force Cryptanalysis of Caesar Cipher

	PHHW	PH	DIWHU	WKH	WRJD	SDUWB
KEY						
1	oggv	og	chvgt	vjg	vqic	rectva
2	nffu	nf	bgufs	uif	uphb	qbsuz
3	meet	me	after	the	toga	party
4	ldds	ld	zesdq	sgd	snfz	ozqsx
5	kccr	kc	ydrpc	rfe	rmey	nyprw
6	jbbq	jb	xcqbo	geb	qldx	mxoqv
7	iaap	ia	wbpan	pda	pkcw	lwnpu
8	hzzo	hz	vaozm	ocz	objv	kvmot
9	gyyn	gy	uznyl	nby	niau	julns
10	fxxm	fx	tymxk	max	mhzt	itkmr
11	ewwl	ew	sxlwj	lzw	lgys	hsjlg
12	dvvk	dv	rwkvi	kyv	kfxr	grikp
13	cuuj	cu	qvjuh	jxu	jewq	fqhjo
14	btti	bt	puitg	iwt	idvp	epgin
15	assh	as	othsf	hvs	hcuo	dofhm
16	zrrg	zr	nsgre	gur	gbtn	cnegl
17	yqqf	yq	mrfqd	ftq	fasm	bmdfk
18	xppe	xp	lqepc	esp	ezrl	alcej
19	wood	wo	kpdob	dro	dyqk	zkbdi
20	vnnc	vn	jocna	cqn	cxpj	yjach
21	ummb	um	inbmz	bpm	bwoi	xizbg
22	tlla	tl	hmaly	aol	avnh	whyaf
23	skkz	sk	glzkx	znk	zung	vgxze
24	rjjy	rj	fkyjw	ymj	ytlf	ufwyd
25	qiix	qi	ejxiv	xli	xske	tevxc

Monoalphabetic Cipher

- rather than just shifting the alphabet
- could shuffle (jumble) the letters arbitrarily
- each plaintext letter maps to a different random ciphertext letter
- hence key is 26 letters long

Plain: abcdefghijklmnopqrstuvwxyz

Cipher: DKVQFIBJWPESCXHTMYAUOLRGZN

Plaintext: ifwewishtoreplaceletters

Ciphertext: WIRFRWAJUHYFTSDVFSFUUFYA

Monoalphabetic Cipher Security

- now have a total of $26! = 4 \times 10^{26}$ keys
- with so many keys, might think is secure
- but would be **!!!WRONG!!!**
- problem is language characteristics

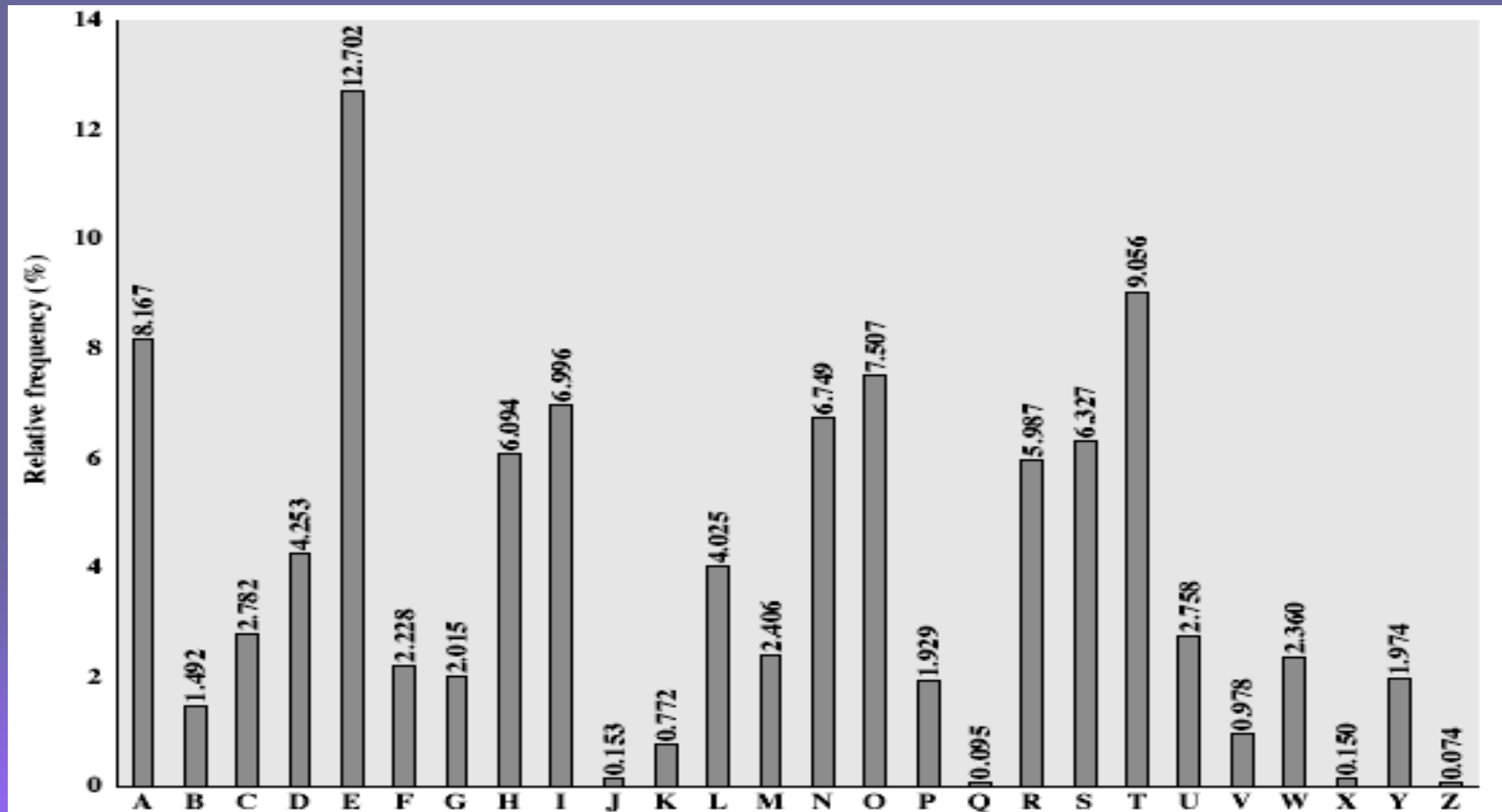


Language Redundancy and Cryptanalysis

- human languages are **redundant**
- eg "th lrd s m shphrd shll nt wnt"
- letters are not equally commonly used
- in English E is by far the most common letter
 - followed by T,R,N,I,O,A,S
- other letters like Z,J,K,Q,X are fairly rare
- have tables of single, double & triple letter frequencies for various languages



English Letter Frequencies



Use in Cryptanalysis

- key concept - monoalphabetic substitution ciphers do not change relative letter frequencies
- discovered by Arabian scientists in 9th century
- calculate letter frequencies for ciphertext
- compare counts/plots against known values
- if caesar cipher look for common peaks/troughs
 - peaks at: A-E-I triple, NO pair, RST triple
 - troughs at: JK, X-Z
- for monoalphabetic must identify each letter
 - tables of common double/triple letters help

Example Cryptanalysis

- given ciphertext:

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFPUPZHMDJUDTMOHMQ

- count relative letter frequencies (see text)

- guess P & Z are e and t

- guess ZW is th and hence ZWP is the

- proceeding with trial and error finally get:

it was disclosed yesterday that several informal but
direct contacts have been made with political
representatives of the viet cong in moscow



Playfair Cipher

- not even the large number of keys in a monoalphabetic cipher provides security
- one approach to improving security was to encrypt multiple letters
- the **Playfair Cipher** is an example
- invented by Charles Wheatstone in 1854, but named after his friend Baron Playfair

Playfair Key Matrix

- a 5X5 matrix of letters based on a keyword
- fill in letters of keyword (sans duplicates)
- fill rest of matrix with other letters
- eg. using the keyword MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Encryption Using Playfair

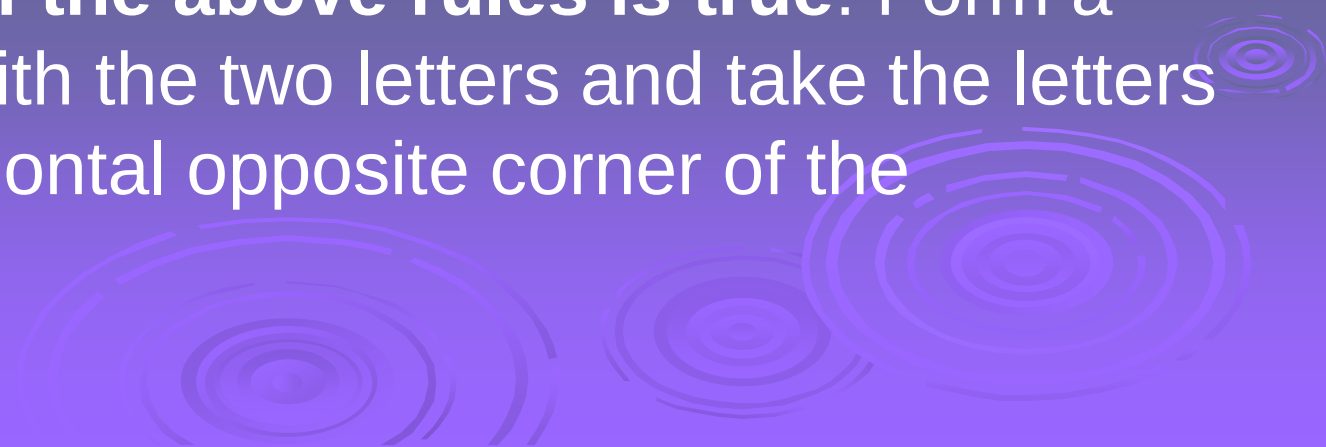
- plaintext is encrypted two letters at a time
 1. if a pair is a **repeated letter**, insert filler like 'X'
 2. if both letters fall in the **same row**, replace each with letter to their immediate right if the original pair is on the right side of the row then wrapping around to the left side of the row happens.
 3. if both letters fall in the **same column**, replace each with the letter immediately below it, if the original pair is at the bottom of the row then again wrapping to top from bottom happens.
 4. If the alphabets are **not in the same row or column** replace them with the alphabets in the same row respectively. But at the other pair of the rectangle defined by the original pair.

Rules for encryption using Playfair Cipher

1. Digrams.
2. Repeating Letters - Filler letter.
3. Same Column | $\downarrow$ | Wrap around.
4. Same row | $\rightarrow$ | Wrap around.
5. Rectangle | $\leftrightarrow$ | Swap



Decryption using playfair.

- **If both the letters are in the same row:** Take the letter to the left of each one (going back to the rightmost if at the leftmost position).
 - **If both the letters are in the same column:** Take the letter above each one (going back to the bottom if at the top).
 - **If neither of the above rules is true:** Form a rectangle with the two letters and take the letters on the horizontal opposite corner of the rectangle.
- 
- The bottom of the slide features several decorative concentric circles in a lighter shade of purple, resembling ripples in water, positioned behind the text of the third rule.

Lets Solve

- Message = MY NAME IS ATUL
- Keyword= PLAYFAIR EXAMPLE
- Encrypt the message with the keyword using Play fair substitution Cipher.



Security of Playfair Cipher

- security much improved over monoalphabetic
- since have $26 \times 26 = 676$ digrams
- would need a 676 entry frequency table to analyse (verses 26 for a monoalphabetic)
- and correspondingly more ciphertext
- was widely used for many years
 - eg. by US & British military in WW1
- it **can** be broken, given a few hundred letters
- since still has much of plaintext structure

Polyalphabetic Ciphers

- **polyalphabetic substitution ciphers**
- Improve on mono alphabetic cipher
- improve security using multiple cipher alphabets
- make cryptanalysis harder with more alphabets to guess and flatter frequency distribution
- use a key to select which alphabet is used for each letter of the message
- use each alphabet in turn
- repeat from start after end of key is reached

Vigenère Cipher

- simplest polyalphabetic substitution cipher
- effectively multiple caesar ciphers
- $C_i = (P_i + K_i) \bmod 26$
- key is multiple letters long $K = k_1 k_2 \dots k_d$
- i^{th} letter specifies i^{th} alphabet to use
- use each alphabet in turn
- repeat from start after d letters in message
- decryption simply works in reverse

Example of Vigenère Cipher

- write the plaintext out
- write the keyword repeated above it
- use each key letter as a caesar cipher key
- encrypt the corresponding plaintext letter
- eg using keyword *deceptive*

key: deceptivedeceptivedeceptive

plaintext: wearediscoveredsaveyourself

ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

Example

Vigenere CIPHER

Key : deceptivedeceptivedeceptive

Plaintext : wearediscoveredsaveyourself

Vigenere Cipher

Key : deceptivedeceptivedeceptive

Plaintext : wearediscoveredsaveyourself

Ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

Key														
PT														
CT														

Key														
PT														
CT														

Vigenere Cipher

★ It consists of the 26 Caesar ciphers with shifts of 0 through 25.

Encryption process:

$$C_i = (P_i + K_{i \bmod m}) \bmod 26$$

Decryption process:

$$P_i = (C_i - K_{i \bmod m}) \bmod 26$$

Vigenere Cipher

Key : deceptivedeceptivedeceptive

Plaintext : wearediscoveredsaveyourself

Ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

Key	3	4	2	4	15	19	8	21	4	3	4	2	4
PT	22	4	0	17	4	3	8	18	2	14	21	4	17
CT	25	8	2	21	19	22	16	13	6	17	25	6	21

Key	15	19	8	21	4	3	4	2	4	15	19	8	21	4
PT	4	3	18	0	21	4	24	14	20	17	18	4	11	5
CT	19	22	0	21	25	7	2	16	24	6	11	12	6	4

Vigenere Cipher – Cryptanalysis

- ★ Determining the length of the keyword.
- ★ Key and the plaintext share the same frequency distribution of letters, a statistical technique can be applied.



Aids

- simple aids can assist with en/decryption
- a **Saint-Cyr Slide** is a simple manual aid
 - a slide with repeated alphabet
 - line up plaintext 'A' with key letter, eg 'C'
 - then read off any mapping for key letter
- can bend round into a **cipher disk**
- or expand into a **Vigenère Tableau**

Security of Vigenère Ciphers

- have multiple ciphertext letters for each plaintext letter
- hence letter frequencies are obscured
- but not totally lost
- start with letter frequencies
 - see if look monoalphabetic or not
- if not, then need to determine number of alphabets, since then can attach each

Hill Cipher

- ★ Multi-letter cipher.
- ★ Developed by Lester Hill in 1929.
- ★ Encrypts a group of letters : digraph, trigraph or polygraph.

- ★ Review few terminologies from linear algebra.
- ★ Concepts to be known:
 - Matrix arithmetic modulo 26.
 - Square matrix.
 - Determinant.
 - Multiplicative inverse.

Hill Cipher

This can be expressed as

$$C = E(K,P) = P \times K \bmod 26$$

$$P = D(K,C) = C K^{-1} \bmod 26 = P \times K \times K^{-1} \bmod 26$$

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26 \quad \leftarrow \text{Encryption}$$

$$C_1 = (P_1 K_{11} + P_2 K_{21} + P_3 K_{31}) \bmod 26$$

$$C_2 = (P_1 K_{12} + P_2 K_{22} + P_3 K_{32}) \bmod 26$$

$$C_3 = (P_1 K_{13} + P_2 K_{23} + P_3 K_{33}) \bmod 26$$

Hill Cipher

Question: Encrypt "pay more money" using Hill cipher with key

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Solution:

p	a	y	m	o	r	e	m	o	n	e	y
15	0	24	12	14	17	4	12	14	13	4	24

Key = 3 x 3 matrix.

PT = pay mor emo ney

Encrypting: mor

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$

↓

$$(C_1 \ C_2 \ C_3) = (12 \ 14 \ 17) \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26$$

$$= (12 \times 17 + 14 \times 21 + 17 \times 2 \quad 12 \times 17 + 14 \times 18 + 17 \times 2 \quad 12 \times 5 + 14 \times 21 + 17 \times 19) \bmod 26$$

$$= (532 \ 490 \ 677) \bmod 26$$

$$= (12 \ 22 \ 1)$$

$$= (M \ W \ B)$$

PT	p	a	y	m	o	r	e	m	o	n	e	y
CT	R	R	L	M	W	B	K	A	S			

Encrypting: ney

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$

$$(C_1 \ C_2 \ C_3) = (13 \ 4 \ 24) \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26$$

$$= (13 \times 17 + 4 \times 21 + 24 \times 2 \quad 13 \times 17 + 4 \times 18 + 24 \times 2 \quad 13 \times 5 + 4 \times 21 + 24 \times 19) \bmod 26$$

$$= (348 \ 312 \ 538) \bmod 26$$

$$= (15 \ 3 \ 7)$$

$$= (P \ D \ H)$$

PT	p	a	y	m	o	r	e	m	o	n	e	y
CT	R	R	L	M	W	B	K	A	S	P	D	H

Hill Cipher

Question: Encrypt "pay more money" using Hill cipher with key

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Plaintext : pay more money

Ciphertext : RRLMWBKAŞPDH

Kasiski Method

- method developed by Babbage / Kasiski
- repetitions in ciphertext give clues to period
- so find same plaintext an exact period apart
- which results in the same ciphertext
- of course, could also be random fluke
- eg repeated “VTW” in previous example
- suggests size of 3 or 9
- then attack each monoalphabetic cipher individually using same techniques as before

Autokey Cipher

- ideally want a key as long as the message
- Vigenère proposed the **autokey** cipher
- with keyword is prefixed to message as key
- knowing keyword can recover the first few letters
- use these in turn on the rest of the message
- but still have frequency characteristics to attack
- eg. given key *deceptive*

key: deceptivewearediscoveredsav

plaintext: wearediscoveredsaveyourself

ciphertext: ZICVTWQNGKZEIIGASXSTSLVWLA

Autokey system

- ★ The periodic nature of the keyword can be eliminated by using a non-repeating keyword that is as long as the message itself.
- ★ Vigenère proposed autokey system, in which a keyword is concatenated with the plaintext itself to provide a running key.

Example

Key : deceptivewearediscoveredsav

Plaintext : wearediscoveredsaveyourself

Ciphertext : ZICVTWQNGKZEIIGASXSTSLVWLA



One-Time Pad

- Also called Vernam Cipher
- if a truly random key as long as the message is used, the cipher will be secure
- called a One-Time pad
- is unbreakable since ciphertext bears no statistical relationship to the plaintext
- since for **any plaintext** & **any ciphertext** there exists a key mapping one to other
- can only use the key **once** though
- problems in generation & safe distribution of key

Transposition Ciphers

- now consider classical **transposition** or **permutation** ciphers
- these hide the message by rearranging the letter order
- without altering the actual letters used
- can recognise these since have the same frequency distribution as the original text

Rail Fence cipher

- write message letters out diagonally over a number of rows
- then read off cipher row by row
- eg. write message out as:

m e m a t r h t g p r y
e t e f e t e o a a t

- giving ciphertext

MEMATRHTGPRYETEFETEOAAT



Row Transposition Ciphers

- a more complex transposition
- Also called simple columnar
- write letters of message out in rows over a specified number of columns
- then reorder the columns according to some key before reading off the rows

Key: 3 4 2 1 5 6 7

Plaintext: a t t a c k p

o s t p o n e

d u n t i l t

w o a m x y z

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

Product Ciphers

- ciphers using substitutions or transpositions are not secure because of language characteristics
- hence consider using several ciphers in succession to make harder, but:
 - two substitutions make a more complex substitution
 - two transpositions make more complex transposition
 - but a substitution followed by a transposition makes a new much harder cipher
- this is bridge from classical to modern ciphers



Rotor Machines

- before modern ciphers, rotor machines were most common complex ciphers in use
- widely used in WW2
 - German Enigma, Allied Hagelin, Japanese Purple
- implemented a very complex, varying substitution cipher
- used a series of cylinders, each giving one substitution, which rotated and changed after each letter was encrypted
- with 3 cylinders have $26^3=17576$ alphabets

Hagelin Rotor Machine



Steganography

- an alternative to encryption
- hides existence of message
 - using only a subset of letters/words in a longer message marked in some way
 - using invisible ink
 - hiding in LSB in graphic image or sound file
- has drawbacks
 - high overhead to hide relatively few info bits

Summary

➤ have considered:

- classical cipher techniques and terminology
- monoalphabetic substitution ciphers
- cryptanalysis using letter frequencies
- Playfair cipher
- polyalphabetic ciphers
- transposition ciphers
- product ciphers and rotor machines
- stenography